

Comparison of the effects of large-grained and nano-sized biochar, ferrihydrite, and complexes thereof on Cd and As in a contaminated soil–plant system

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ABSTRACT

Cd and As are difficult to co-remediate in co-contaminated soils. In this study, remediation materials comprising large-grained and nano-sized biochar (BC), ferrihydrite (FH), and complexes thereof were added to Cd- and As-contaminated soil. The uptake of Cd and As by pak choi (*Brassica chinensis* L.) was then evaluated using a pot experiment and the Cd and As concentrations of the soil pore water and leaching water were measured. The Cd and As concentrations of the pore and leaching water were slightly increased with the addition of BC, and decreased with addition of FH and the biochar–ferrihydrite complex (BC-FH). However, nano-sized BC (BC_N), FH (FH_N), and BC-FH (BC-FH_N) had little influence on the decreases in Cd and As of the two monitored water types. Large-grained remediation materials, rather than nanomaterials, decreased the Cd and As concentrations of the two monitored water types. Nonetheless, nanomaterial treatments more effectively decreased the Cd and As concentrations in plants by an average of >10% relative to the large-grained treatments. The DLVO theory analysis suggested that BC_N, FH_N, and BC-FH_N immobilized in the topsoil, adsorbed heavy metals in the rhizosphere soil. The remainder of the nano-sized materials was dispersed in the rhizosphere soil pores, shielding the uptake of Cd and As by the roots. Although the doses of nanomaterials used in this study were less than one-fortieth of those of the large-grained materials, changes in the plant rhizosphere microenvironment caused by the nanomaterials decreased the risk of toxicity transfer from the soil to the plants.

1. Introduction

Cd and As are toxic metals/metal(loid)s that can enter and accumulate in soil and the environment through industrial waste, atmospheric emissions, mining, agricultural applications, and other anthropologic activities (Alam et al., 2003; Camm et al., 2004; de Meeûs et al., 2002; Horn et al., 2003). Both metal(loid)s are highly reactive in the natural environment. The excessive accumulation of heavy metal(loid)s in agricultural soil may cause heavy metal(loid)s accumulation by crops, which, through the food chain, would be hazardous to human health (Kubier et al., 2019). Exposure to Cd can cause kidney and liver

dysfunction, weaken the immune system, and adversely affect reproduction (Chaney, 2015), while As exposure can lead to cancers, vascular disease, neurological disorders, and skin lesions, among others (Naujokas et al., 2013). These two elements can co-exist in soils, resulting in a co-contamination by Cd and As, which has become a major concern in agricultural ecosystems (Wu et al., 2020). Therefore, the remediation of agricultural soil that has been co-contaminated by Cd and As is crucial. However, the distinct charge of Cd and As in soil produce different geochemical behavior (Qiao et al., 2018; Tian et al., 2021). Qiao et al. (2018) found that Cd and As bioavailability show opposite change with pH increase. For example, during alteration of wetting and drying, Cd

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bioavailability may decrease under flooded conditions due to precipitation with sulfide, but As bioavailability may increase because of As adsorbed to Fe(III) oxides released into solution (Honma et al., 2016; Yu et al., 2017). The contrasting chemical behaviors of Cd and As make it difficult for the simultaneous stabilization and the remediation materials should be carefully selected to immobilize both Cd and As in co-contaminated soil.

Various soil remediation materials, including biochar (BC), compost, organic fertilizers, lime, and metal (Fe, Mn, Al) oxides have been utilized to effectively immobilize residual heavy metals in the soil and restore soil fertility (Guo et al., 2020; Kaurin et al., 2018; Rehman et al., 2020). For example, BC is an environmentally friendly and economical material that has been widely used as a soil amendment to improve soil quality (Peng et al., 2019). BC can be capable of binding cations due to its large and negatively charged surface (Huang et al., 2020). When it is applied to soil, it predominantly reduced Cd mobility by ion exchange, electrostatic attraction, and functional groups (e.g., $-\text{OH}$, $-\text{COOH}$, $-\text{C}=\text{O}$, and $\text{C}=\text{N}$) (He et al., 2019). However, BC lacks anion exchange capacity owing to its negatively charged surface (Lawrinenko and Laird, 2015). Hence, BC has exhibited a limited efficiency for remediating anions such as As (Beesley et al., 2013). For example, uptake of As to plant was reduced by BC addition to the soil, but solubility and mobility were increased in soil (Tian et al., 2021). For Cd and As co-contaminated soil, the application of BC can effectively immobilize Cd but might increase the bioavailability of As. Compared to BC, iron oxide surfaces had a high adsorption affinity for As (Jiang et al., 2013). Among the iron oxides, ferrihydrite (FH) has the highest adsorption capacity for As of ~ 2.1 mmol/g (Dixit and Hering, 2003) because of an extremely high surface area ($180\text{--}800$ m²/g) and exposed abundant adsorption sites (Gustafsson et al., 2009; Raven et al., 1998). FH surface groups can induce ligand exchanges with oxyanions (for example, arsenate AsO_4^{3-}) to form stable $\equiv\text{Fe}-\text{O}-\text{As}$ species on the surface (Fendorf et al., 1997; Guo et al., 2013). Therefore, FH is valuable as a modifier to stabilize As in the soil. Combined remediation materials have been shown to have superposition passivation effects toward reducing the heavy metal availability of soil relative to single remediation materials (Friesl-Hanl et al., 2009). In order to simultaneously stabilize Cd and As, combined remediation materials, such as Ca-based magnetic BC, have previously been applied during soil remediation (Wu et al., 2020). Thus, for co-contaminated by Cd and As, it seems that a combined remediation material may be a good choice for treating co-contaminated soil.

Recently, it has been reported that nanomaterials (1–1000 nm) can be used to remediate soil and sediment contaminated by heavy metals (Huang et al., 2018a, 2018b). For example, Song et al. (2017) demonstrated that the application of multi-walled carbon nanotubes decreased Cd concentrations released from sediments. Gil-Diaz et al. (2016) found that nanoscale zero-valent Fe addition could not completely prevent plant absorption of As, but did reduce the availability of As in the soil, thus reducing the toxicity level. Besides, researchers have reported that nanomaterials have higher reactivities and adsorption abilities than larger-sized grains of the same material (He et al., 2013; Zhang et al., 2010). For example, Liu and Zhao (2007) found that Fe phosphate nanoparticles had better performance than larger particles when immobilizing Cu(II) in soil. However, these soil amendments mainly utilized traditional bulk particles, rather than nanoscale materials, in the immobilization of soil co-contaminated by Cd and As.

Natural organic matter can influence the immobilization and transport of nanoparticle-associated contaminants (Hammes et al., 2013). Humic acid (HA) is a stable component of organic matter in soil (Hemati et al., 2012) and can be easily adsorbed on many solid surfaces, including those of BC and metal oxide nanoparticles (Chen et al., 2007; Wang et al., 2013). HA can directly adsorb heavy metal ions (e.g., Cd^{2+}) and enhance heavy metal adsorption on Fe oxyhydroxide colloids when HA is also adsorbed on these oxides (Abasiyan and Tofighi, 2013; Arias et al., 2002). Moreover, As removal by Fe oxide minerals is inhibited by HA through the formation of negatively charged surface sites or the

blocking of adsorption sites on the adsorbent (Li et al., 2017; Weng et al., 2009). Previous studies have shown that As adsorption on hydrous ferric oxides decreased from over 95%–10% in the presence of HA at a mass concentration of 25 times that of As (Du et al., 2014). Thus, HA is an important component in regulating the immobilization and transport of contaminants and colloids (Wang et al., 2012).

Overall, data regarding the application of larger and nano-sized BC and FH to soil co-contaminated with As and Cd is lacking, especially in the presence of HA. Therefore, this study compared the effects of large-grained and nano-sized BC, FH, and complexes thereof on the stabilization of Cd and As in co-contaminated soil in the presence and absence of HA. In addition, we determined the effect of the dispersal of the nano-sized materials on Cd and As uptake by pak choi. This study illustrates the effects of a nano-sized remediation materials on decreases in Cd and As bioavailability and plant bioaccumulation, and provides a guide for selecting appropriate remediation approaches for heavy metal-contaminated soil.

2. Materials and methods

2.1. Experimental materials

The experimental soil was collected from the top 20 cm layer of a wheat field surrounding an abandoned Cu mine in the village of Baoding, Hebei Province, China ($115^{\circ}44'05''\text{E}$, $38^{\circ}47'18''\text{N}$). After air-drying, the soil was passed through a 0.83-mm sieve. The total Cd and As concentrations of the soil were 4.36 and 28.7 mg kg⁻¹, respectively; these significantly exceed the standards for Cd and As concentrations in soil used for agricultural production in China (GB15618-2018). Besides, the total Cu and Zn concentrations also significantly exceed the standards for their concentrations in soil, but they are little harmful to plants compared with Cd and As. Thus, we mainly discussed the effect of the materials on Cd and As in this study. The physical and chemical properties of the soil are presented in Table S1.

The BC was prepared by the pyrolysis of wheat straw at 300 °C for 2 h in an N₂ atmosphere. The collected BC sample was cooled to room temperature in an N₂ atmosphere and then was sonicated in a 6 M HCl solution to remove acid-soluble oxides. After an acid wash, the BC sample was repeatedly washed with ultrapure water until the conductivity of the filtrate was 5.0 μS/cm. The dry BC was ground into a powder, passed through a 0.45-mm nylon sieve, and labeled as BC. The properties of the BC were characterized in our previous study (Qian et al., 2020). The pH of the BC was 10.1 and the pore sizes distributions ranged from 10 μm to 100 μm, with a specific surface area of 2.1 m² g⁻¹. The FH was synthesized by titrating dissolved Fe(NO₃)₃·9H₂O (Sino-pharm Chemical Reagent Co. Ltd.) with NaOH to pH 7.5 (Jia et al., 2007; Qian et al., 2020). The synthesized FH was verified using its X-ray diffraction pattern in a previous study (Ma et al., 2018a), and the specific surface area was 277.9 m² g⁻¹ (Ma et al., 2018c). Suspensions of BC_N and FH_N were prepared by adding 0.4 g of BC and FH, respectively, to 400 ml of Milli-Q water, which was vigorously stirred and sonicated for 60 min. After allowing the suspension to settle for 24 h, it was dialyzed by siphoning and used immediately after preparation. The concentrations of BC_N and FH_N were measured using a total organic carbon (TOC) analyzer (MODEL 1030S, OI Analytical) and atomic absorption spectrometry (AAS, AAnalyst900T, PerkinElmer), respectively. The HA used in this study was purified from commercial HA following the alkali/acid fractionation procedure (Valdrihi et al., 1996).

2.2. Pot experiment

A pak choi (*Brassica chinensis* L.) pot experiment was conducted to compare the effects of the different remediation materials on the fate of heavy metal(loid)s in the soil–plant system. The contaminated soil was classified into two types of treatments: (1) contaminated soil was artificially mixed with large-grained BC, FH, and complexes thereof (BC-

FH); (2) contaminated soil received additions of nano-sized particle solutions (BC_N , FH_N , and $BC-FH_N$). The 10 different soil groups and additions are shown in Table 1, and the experimental design is shown in Figure S1. One pore water sampler was carefully inserted into a pre-bored pilot hole in each pot to collect pore water. Leaching water was collected from the bottom of each pot. After every irrigation, the leachates were collected over a 24-h period using a big pot placed beneath each experimental pot, and then filtered with Whatman paper (grade 40, 125 mm diameter). The pore water and leaching water were collected three times (1, 20, and 48 d) and seven times (1, 8, 16, 24, 32, 40, and 48 d), respectively, over 48 d to evaluate the influence of the remediation materials on the dynamic transfer of heavy metal(loid)s in the soil.

After 48 days of treatment, the pak choi seeds were rinsed three times with ultrapure water and cultivated in the experimental soil in a controlled environment (23 °C, 33% relative humidity) with a 16 h per day photoperiod for 20 days. All pot treatments were conducted in triplicate and each pot contained six seedlings. After 20 days, sampling plants were harvested for contents analysis of Cd and As. Meanwhile, the surface soils after harvest of the plants were also collected by stratified sampling (0–5, 5–10, and 10–15 cm, upper, middle, and under) to determine the partitioning of Cd, Fe and As in the soil.

2.3. Analytical methods

Before analysis, the soil samples were freeze-dried, ground to produce a fine powder, sieved (<0.15 mm), and digested using $HNO_3-H_2O_2$ (USEPA, 1996). The plant material was digested with HNO_3-HClO_4 (5:1). As concentrations were measured using atomic fluorescence spectrometry (AFS, PSA-10.825, PS Analytical), and Cd and Fe concentrations were measured using atomic absorption spectrometry (AAS, PinAAcle 900T, PerkinElmer). The elemental recoveries in the reference samples (certified soil GBW07307, Institute of Geophysical and Geochemical Exploration, Langfang, China) were in the range of 90.9 ± 3.5 to $97.4 \pm 4.9\%$. The pH values of the soil samples in a suspension containing 10 mM of $CaCl_2$ (solid-to-liquid ratio of 1:5) were measured using a pH/oxidation reduction potential meter. Digestion in HCl was used to remove all inorganic carbon from the soil samples, and the TOC concentration was determined using a TOC analyzer.

The As sequential extraction method (Wenzel et al., 2001) was adopted to investigate the As fractions in the soil samples. The details of this extraction method are presented in Table S2. The BCR sequential extraction method, a simple three-stage procedure, was used to determine the Cd and Fe fractions of the soil samples (Arain et al., 2008; Sun et al., 2019). The details of the BCR sequential extraction method are presented in Table S3. Three replicate samples were analyzed for Cd, As, and Fe. The supernatants collected from the BCR and As sequential extractions were filtered through 0.45- μ m membrane filters before

analysis.

2.4. DLVO theory

The Derjaguin-Landau-Verwey-Overbeek (DLVO) theory was used to qualitatively understand the interaction energy between colloid and the soil (sphere-plate) through calculating the total particle-soil interaction energy as the sum of Lifshitz-van der Waals (LW) and electrical double layer (EDL) interactions (Derjaguin and Landau, 1941; Verwey and Overbeek, 1947). It also can be used to understand the nanoparticles in soil pore. A series of data (e.g., zeta potential, the radius, ionic strength) were used as input of the DLVO theory calculation. Details of DLVO theory and operative equations are presented in Supplementary material 5.

2.5. Statistical analyses

All values reported represent the means of three independent replicate samples. Statistical analyses were performed using SPSS 17.0 software and the figures were generated using Origin 2018. An analysis of variance (ANOVA) and a subsequent Duncan test were performed at a statistical significance level of $p < 0.05$.

3. Results and discussion

3.1. Effects of remediation materials on Cd and As in the pore water

In order to analyze the release and immobilization of Cd and As from the soil, pore water monitoring was conducted. Fig. 1 illustrates the changes in Cd and As concentrations in the pore water over 48 days following the application of the remediation materials to the contaminated soil. On the first day, high Cd and As concentrations in the pore water were observed with all treatments. For Cd, except for the BC_N treatment, the treatments had lower concentrations (10–51%) than that in the control (Fig. 1a). Among all the treatments, those with additions of FH and $BC-FH$ had the lowest Cd concentrations, indicating good Cd immobilization in the soil. For As, all the treatments had lower concentrations (3–24%) than that in the control (Fig. 1b). Except for the BC_N treatment, the treatments with added remediation materials reduced the Cd and As concentrations compared to those in the control. In addition, large-grained remediation treatments had lower Cd concentrations (average 17%) than those of the nanomaterial treatments, while no differences were observed in the As concentrations. As the experiment continued, the Cd and As concentrations were decreased by 41–61% and 50–90%, respectively, after 20 days, compared to the first day. After 48 days, the Cd and As concentrations were decreased by 50–65% and 74–90%, respectively, compared to their initial concentrations. The decreased Cd and As concentrations may have been due to immobilization by the remediation materials and leaching losses.

On day 48, the FH and $BC-FH$ treatments had reduced the Cd (by 32% and 39%, respectively) and As (by 61% and 48%, respectively) concentrations in the pore water compared to those in the control, while the FH_N and $BC-FH_N$ treatments had caused no evident differences in the Cd concentrations and decreases in the As concentrations (by 21% and 24%, respectively). Moreover, compared to the control, BC and BC_N application slightly increased the concentration of Cd and As (about 10%) in pore water, but the difference was not significant. Overall, the large-grained treatments (BC , FH , and $BC-FH$) yielded Cd and As concentrations about 26% lower on average than those of the nanomaterial treatments (BC_N , FH_N , and $BC-FH_N$). This indicates that the large-grained remediation materials lowered the Cd and As concentrations more effectively than the nanomaterials.

3.2. Effects of remediation materials on Cd and As in the leaching water

Fig. 2 and Tables S4-S5 illustrate the changes in Cd and As

Table 1
Different remediation material treatments in the pot experiment.

Group	Solid additive	Liquid additive (50 ml per time and total seven times) 50 ml is roughly equal to 1/3 pore volume of the soil in the pot	Label
1	300 g soil	H_2O	Control
2	300 g soil, 1.5g BC	H_2O	BC
3	300 g soil, 1.5g FH	H_2O	FH
4	300 g soil, 1.5g $BC-1.5g FH$	H_2O	$BC-FH$
5	300 g soil	100 mg L^{-1} BC_N	BC_N
6	300 g soil	100 mg L^{-1} BC_N -10 mg L^{-1} HA	BC_N-HA
7	300 g soil	100 mg L^{-1} FH_N	FH_N
8	300 g soil	100 mg L^{-1} FH_N -10 mg L^{-1} HA	FH_N-HA
9	300 g soil	50 mg L^{-1} BC_N -50 mg L^{-1} FH_N	$BC-FH_N$
10	300 g soil	50 mg L^{-1} BC_N -50 mg L^{-1} FH_N -10 mg L^{-1} HA	$BC-FH_N-HA$

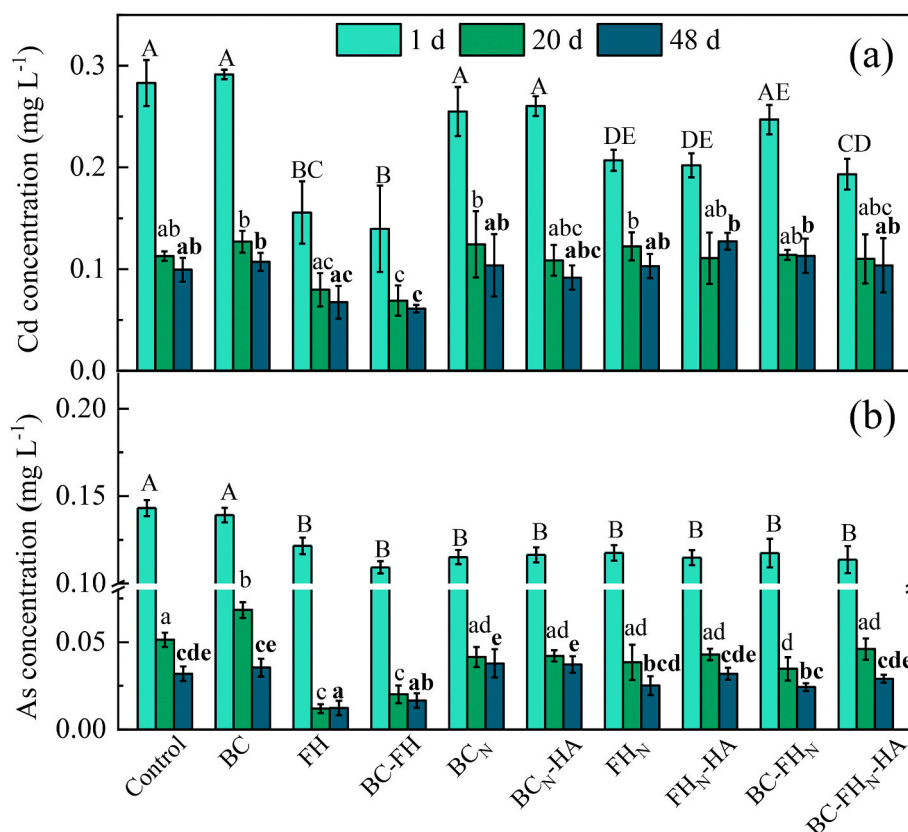


Fig. 1. Influences of different treatments on soil pore water concentrations of Cd (a) and As (b) at different sampling times. Different letters (A, B, C, D, E, and a, b, c, d, e, and bold a, b, c, d, e) represent significant differences ($p < 0.05$) between treatments at the same sampling time.

concentrations in the leaching water over 48 days following the application of the remediation materials; the changes were slightly different from those in the pore water. On the first day, except for the FH and BC-FH treatments, most of the treatments exhibited high Cd and As concentrations in the leaching water. The Cd concentrations in the leaching water were the lowest in the FH and BC-FH treatments, with concentrations that were 65% and 68% lower than that in the control, respectively. These two treatments also had the lowest As concentrations, which were lower than that in the control by 95% and 93%, respectively. The addition of BC yielded the highest As concentration, which was 49% higher than that in the control. Moreover, large-grained remediation materials (BC, FH, and BC-FH) yielded Cd and As concentrations 49% and 41% lower on average, respectively, than those measured in the leaching water for nanomaterial treatments (BC_N, FH_N, and BC-FH_N). The decrease in Cd and As concentrations in the leaching water was less than that in the pore water (Figs. 1 and 2). This was attributed to the greater adsorption of Cd and As by the large-grained materials during the leaching process (Tables S4 and S5).

Overall, the leaching water Cd concentration was decreased gradually (Fig. 2a, Table S4). After 24 and 48 days, the leaching water Cd concentrations were decreased by 41–72% and 51–82%, respectively, compared to those on the first day. Similarly, the treatments without FH caused decreases in As concentrations (59–75%) (Fig. 2b, Table S5). However, after 24 days, the As concentrations increased by 25% and 40% under the FH and BC-FH treatments, respectively, via the gradual release of adsorbed As. At the last sampling, As concentrations in the leaching water were increased by 83–93% compared to those on the first day. Although the As concentrations in the FH and BC-FH treatments were increased, the As concentrations of the leaching water with these two treatments were still the lowest of all the treatments.

On day 48, the FH and BC-FH additions decreased the Cd (by 54% and 61%, respectively) and As (by 55% and 45%, respectively) leaching

water concentrations (Fig. 2) compared to the concentrations in the control. This indicates that FH and BC-FH may have significantly affected the control of Cd and As mobility. However, BC application caused a marginal increase in the As concentration (14%) of the leaching water after 48 days compared to that in the control (Fig. 2b). Similar to the pore water concentration behaviors, the large-grained remediation materials (BC, FH, and BC-FH) yielded Cd and As concentrations of 20% and 28% lower on average in the leaching water, respectively, than those yielded by the nanomaterials (BC_N, FH_N, and BC-FH_N). Because of adsorption during the leaching process, the effects of the large-grained remediation materials on the decreases in the Cd and As concentrations in the leaching water were more significant compared to those in the pore water.

There were no significant differences in the Cd and As concentrations in the pore and leaching water for treatments with and without HA (Figs. 1 and 2, Table S4 and S5), indicating that the influence of HA on the remediation process using nano-sized materials was not obvious in this study. In addition, BC and FH were not detected in the leaching water.

For the treatments used in this study, rapid decreases in the Cd and As concentrations were observed for the first eight days of the experiment, followed by more gradual decreases. This could be caused by the early removal of most of the available Cd and As, while later removals addressed more recalcitrant Cd and As. For a prolonged experimental period, further decreases in the leached metal concentrations would be expected. The decreases in Cd and As concentrations in the leaching water may be related to immobilization by the remediation materials and leaching losses (Tables S4 and S5). However, the leached Cd and As percentages were only ~1–4% and ~0.1–0.4%, respectively, compared to the total amounts (Figure S2). Therefore, we speculated that the impact of leaching losses was not significant.

The surfaces of Fe hydrous oxides had better effects on the retention

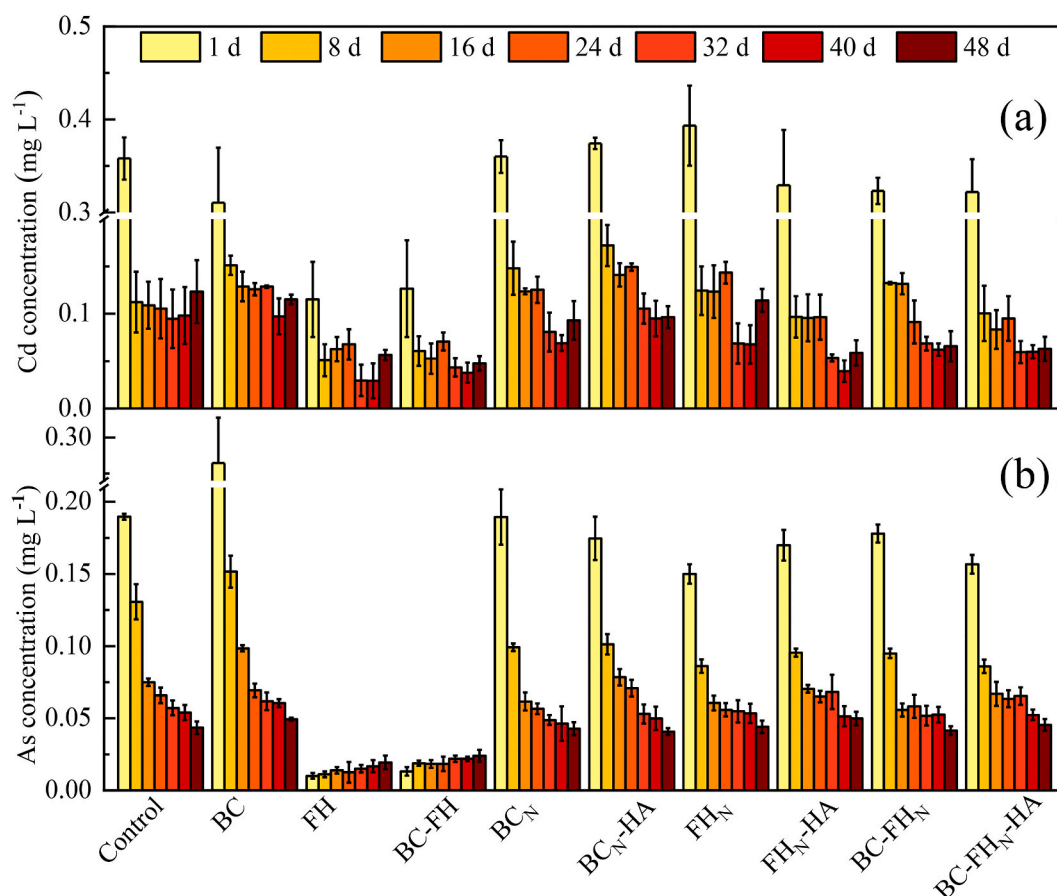


Fig. 2. Influences of different treatments on leaching water concentrations of Cd (a) and As (b) at different sampling times. All statistical information is in the Supplementary Materials (Tables S4 and S5).

of As and Cd from the soil. Mench et al. (2000) found that Cd formed mononuclear complexes on the FH surfaces, and that AsO_4^{3-} formed inner-sphere surface complexes with hydrous ferric oxide. This mechanism allows FH to absorb Cd and As from the soil, resulting in decreased pore water concentrations of Cd and As. BC particles are considered to be good complexation agents for cationic metals in soil solutions (Houben et al., 2013), which can immobilize Cd by ion exchange, electrostatic attraction, and functional groups that reduce Cd mobility, thus influencing its availability (He et al., 2019). Beesley et al. (2010) applied BC to soil contaminated with multiple elements and found that, while pore water Cd and Zn concentrations in the pore water were decreased significantly, those of Cu and As were increased. Van Poucke et al. (2020) also found that BC could reduce the leaching water Cd concentrations; compared to the total concentration, 0.3–0.7% of Cd was leached. However, no significant changes were observed in the Cd and As concentrations after BC application in the present study, which may have been due to the neutral soil pH. It is generally accepted that high pH values induce the immobilization of Cd and the mobilization of As (Fitz and Wenzel, 2002; Hartley et al., 2009). In addition, large-grained remediation materials decreased the pore water Cd and As concentrations more effectively than the nanomaterials did. These results are inconsistent with those reported in previous studies, where nanomaterials generally showed higher reactivity and adsorption abilities than large-grained materials (He et al., 2013; Zhang et al., 2010). This may be related to the higher dosages of the large-grained remediation materials, which were over 40 times the nanomaterial dosages.

Overall, the FH and BC-FH treatments yielded much lower Cd and As pore and leaching water concentrations than the control, which indicated that these treatments may promote Cd and As immobilization and hinder the transport risk of Cd and As from the soil into pore water and

groundwater. Based on the combined pore water results, the large-grained materials may have better effects on decreasing the Cd and As concentrations in pore water and leaching water compared to nano-sized materials.

HA can interact with metal ions to form metal–organic complexes and influence contaminant transport by affecting the adsorption of contaminants onto mineral colloids (Ma et al., 2018b; Tang et al., 2014). For example, HA can inhibit As adsorption on Fe oxide minerals by forming negatively charged surface sites or by blocking the adsorption sites of the adsorbent (Li et al., 2017; Weng et al., 2009). In general, the presence of HA may alter the surface properties, stability, and transportation of nanomaterials, effectively influencing the removal and transportation of heavy metal(loid)s (Tang et al., 2014). However, HA had no obvious influences on the remediation process of the nano-sized materials in this study.

3.3. Changes of pH, Cd and As fractions in soil after remediation material application

After 20 days of growth, the pak choi plants were harvested to analyze their concentrations of Cd and As. The soil was also collected to measure its pH and heavy metal contents. For all remediation materials, the single or mixed BC treatments slightly increased the soil pH compared to that of the control. Similar effects on soil pH were observed in previous studies (Meng et al., 2018; Zhai et al., 2018). However, the differences in pH were small in this study, usually <0.3 units, and all soil pH values were ~7.0 (Figure S3). In addition, no significant changes in the pH of different soil layers were observed within individual treatment.

We also determined the fractions of Cd and As, not including residual

fractions, in the different soil layers with the addition of remediation materials (Fig. 3). However, small differences appeared in the Cd and As fractions between individual treatments and the control. The treatment groups had lower acid-soluble Cd (Cd-F₁) and non-specifically bound As (As-F₁) proportions compared to the control. The influences of the large-grained remediation materials on the decreases in the proportions of Cd and As fractions (3.97% on average) were somewhat greater than those of the nano-sized remediation materials (2.36% on average), indicating better Cd and As immobilization by the large-grained remediation materials. This was consistent with the behaviors of the Cd and As concentrations in the pore and leaching water. In addition, no significant differences were observed in the Cd and As fractions at different soil depths. The Fe-F₂ contents in different profiles increased by ~200 mg kg⁻¹ when large-grained materials were added. Meanwhile, the addition of nano-sized materials only caused increases in the Fe-F₁ and Fe-F₂ contents in the upper and middle layers, thus indicating the retention of FH nanoparticles (Figure S4).

3.4. Uptake of Cd and As by plants

The total Cd and As contents of the pak choi plants are shown in Fig. 4. The overall plant Cd concentrations were decreased by 6–33%, except in those treated with FH, compared to that in the control. For As, all remediation materials except BC significantly decreased the As uptake by the plant (14–59%) compared to that in the control. Compared to the control, the BC-FH_N reduced the total As concentration of the plant by over 59%, which was the most effective, and the total Cd concentration by 22%. The BC-FH_N-HA treatment reduced the As and Cd concentrations in the plant by 51% and 33%, respectively, which was

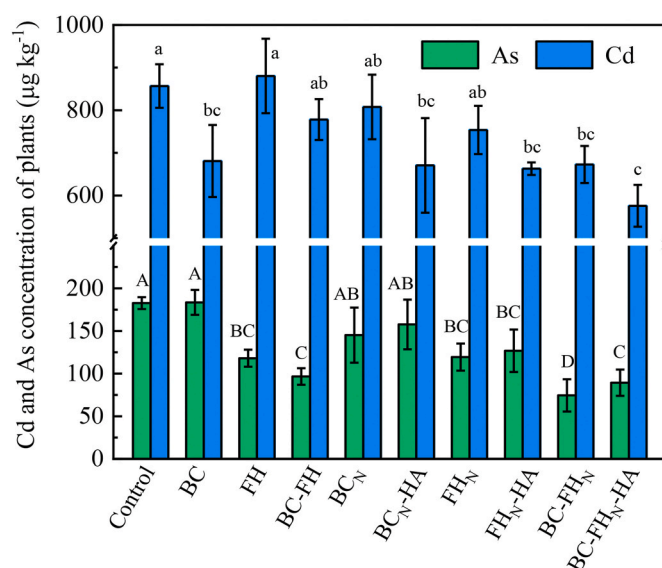


Fig. 4. Influences of different treatments on plant Cd and As concentrations. Different letters (A, B, C, D, E and a, b, c, d, e) represent significant differences ($p < 0.05$) between treatments in same element.

also the most effective for Cd reduction. Furthermore, compared to the control, the FH and BC-FH treatments had good effects on the As concentrations, which were reduced by 35% and 47%, respectively, but no changes occurred in the Cd concentrations in the plants. Overall, the

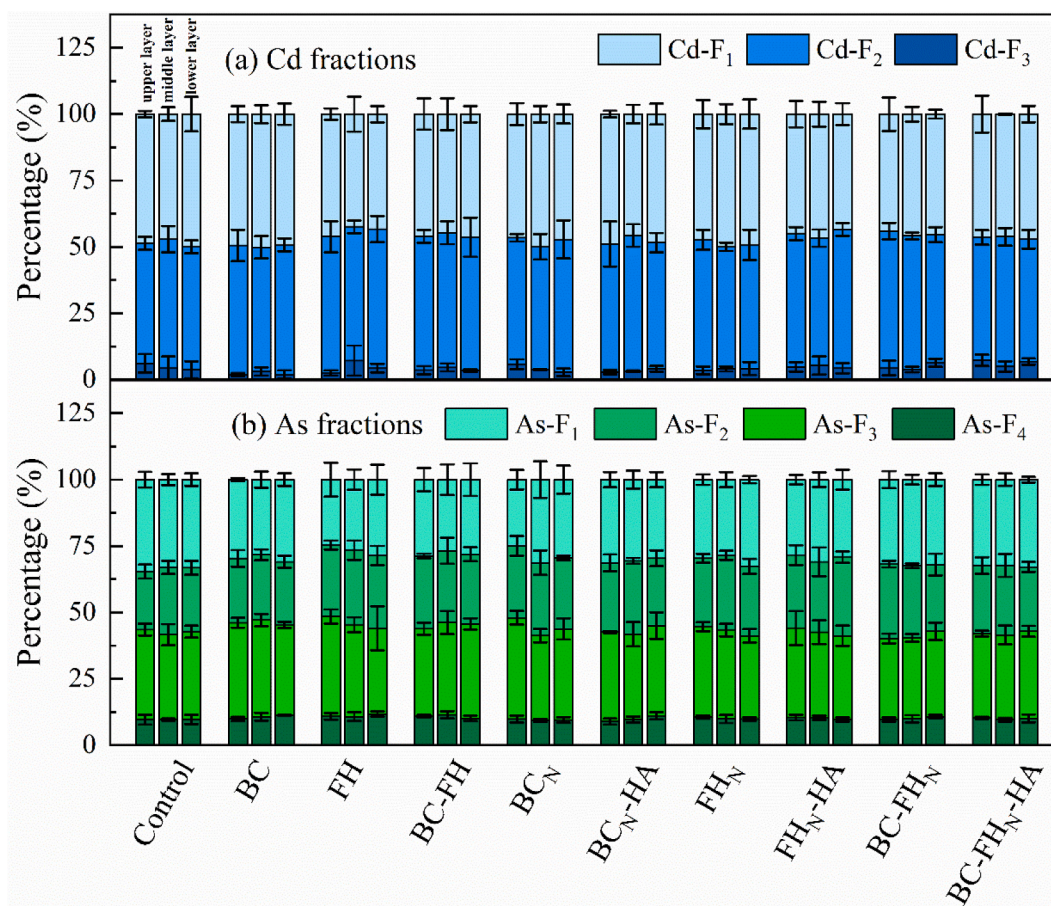


Fig. 3. Influences of different treatments on concentrations of Cd (a) and As (b) fractions in soil profiles. The three columns from left to right for each treatment label represent the upper, middle, and lower layers. There is no significant difference between treatments for all fractions.

large-grained remediation treatments (BC, FH, and BC-FH) had plant Cd and As concentrations over 14% higher on average than the nano-material treatments (BC_N , FH_N , and $BC-FH_N$) did. This indicated that the nano-sized treatments were more effective in reducing the total Cd and As concentrations of the plants, although the application of the nano-sized materials had a smaller influence on the transfer of Cd and As from the pore and leaching water to soil. The reason was that added nano-sized materials mainly were immobilized in the topsoil, where the roots of pak choi were mainly distributed, and little or no leached to the under layers. Consequently, nano-sized materials effectively reduced Cd and As in plants. However, sampling pore water was mix of the soil water in top, middle and under layer and leaching water flowed through whole soil profile, which reflected the average situation of pot rather than top layer alone. Therefore, the influence of nano-sized treatments on pore and leaching water was not significant as compared to control. In further study, we will consider to combine large-sized with nano-sized material treatment for reducing risk of transfer to plants and underwaster. In addition, a significant difference appeared between the plant As concentrations for the $BC-FH_N$ and $BC-FH_N-HA$ treatments, while other treatments had no significant differences on the plant Cd and As concentrations with HA use. This indicated that the influence of HA on the remediation processes using nano-sized materials was not obvious in this study.

3.5. Mechanism of decreases in Cd and As uptake by plants

To further discuss the decreases in Cd and As uptake by the plants as influenced by nanoparticles, the Derjaguin–Landau–Verwey–Overbeek (DLVO) energy between the nanoparticles and soil particle surfaces was calculated. In this study, the zeta potential of the soil was -16.8 mV (Table S6), which is similar to those of soil colloids (between -15.7 and -21.6 mV) in northern China (Chen et al., 2020; Ma et al., 2016). Except for the FH_N (23.7 mV) particles, all the nanoparticles exhibited negative potentials of -39.1 to -43.9 mV. However, the low negative potentials of the soil, the small sizes of the nanoparticles (150.2 – 318.2 nm), and certain ionic concentrations (6.8 – 19.8 mM) (Table S6) produced low primary energy barriers between the nanoparticles and the soil (~ 30 kT) (Fig. 5). In our previous study, when the primary energy barrier was close to 100 kT, only 13% of the soil colloids penetrated the sand column (Ma et al., 2016). Similarly, a previous study reported that 14.6% of the BC_N adsorbed with Cr penetrated a sand column at 5 mM ionic concentration for a 180 -kT primary energy barrier between BC_N and the collector (Jin et al., 2020). Moreover, FH_N adsorbed with HA was completely trapped in the sand column at a primary energy barrier of

~ 200 kT (Ma et al., 2018a). Therefore, a low primary energy barrier of ~ 30 kT could not sustain nanoparticles over long distances. In addition, the mixture of BC_N and FH_N produced many large particles with diameters of 3000 – 6000 nm, which enhanced the filtration effect (Molnar et al., 2015). In this study, the nanoparticles were immobilized in the topsoil under all conditions. The slight increases in the topsoil Fe concentrations and the absence of BC_N and FH_N in the leachate confirmed this immobilization.

The mechanism that produced the decreased plant uptake of Cd and As, regulated by the nano-sized materials, is shown in Fig. 6. The BC_N and FH_N were deposited uniformly on the surfaces of the soil particles, and these nano-sized materials immobilized Cd and As in the root layer more precisely than the large-grained materials mixed throughout the soil layer. Although the added amounts of the nanomaterials were small, the immobilization effect of the nano-sized materials on the heavy metal (loid)s was more prominent in the root layer because of their high specific areas. As the nanoparticles accumulated in the surface soil, the soil pore environment gradually changed. In particular, the negatively charged BC_N and HA could increase the negative charge on the soil surfaces. Although a small change cannot promote the long-distance transport of nanoparticles applied later, here it facilitated their transport downward for a certain distance. The effective depth of the nanoparticles was thereby expanded. In addition, some of the nanoparticles may have remained stable in the surface pore water because of the increase in the negative charge. The sizes of these nanomaterials allowed them to mix more thoroughly with the surface soil compared to the large-grained materials, thus forming barriers around the plant roots that prevented the uptake of dissolved Cd and As from the soil pore water. Furthermore, their surface structures may differ from those of the large-grained materials, causing altered reactivity or varied crystal chemistry. Thus, the nano-sized materials may have higher reactivities and sorption abilities than the large-grained materials (Zhang et al., 2010). This indicates that the nanomaterial treatments were more effective at reducing the total Cd and As concentrations of the plants, despite the applied amounts of the nano-sized materials being just one-fortieth of those of the bulk particle materials.

Different nanomaterials have different remediation effects on soil contaminated with Cd and As. For example, nano-hydroxyapatite particles (nHAP) can remediate Cd by decreasing the concentration of water-soluble Cd in contaminated soils through surface complexation and intraparticle diffusion, thereby reaching low Cd phytoavailability (Cai et al., 2019; He et al., 2013). Gil-Diaz et al. (2016) found that nanoscale zero-valent iron (nZVI) removed the effects of As in soil by decreasing the amounts of the more available fractions of As and increasing the amount of As in the residual fraction, thus further decreasing As uptake. Nanoscale materials can greatly hinder the release of metals to the environment and their uptake by plants by decreasing the active Cd and As fractions, despite their reduced capacity to remove metals from soil (Zhang et al., 2010). In this study, nano-sized materials caused low Cd and As uptakes by pak choi through changing the pore environments in soil. Thus, nanoscale materials (e.g., BC_N and FH_N) are potential solutions for the remediation of agricultural soil co-contaminated by Cd and As and the alleviation of the hazards these species pose to plants and humans.

4. Conclusion

In this study, nano-sized and large-grained BC, FH, and complexes thereof were applied to soil co-contaminated by Cd and As to compare their remediation effects on Cd and As mobility in soil and accumulation in plants. The results indicated that the large-grained BC increased the Cd and As concentrations in the pore water and leaching water, while the large-grained FH and BC-FH decreased the Cd and As concentrations in the pore water and leaching water. However, the BC_N , FH_N , and $BC-FH_N$ treatments had relatively lower efficacy in decreasing the heavy metal contents in the two monitored water types. The large-grained

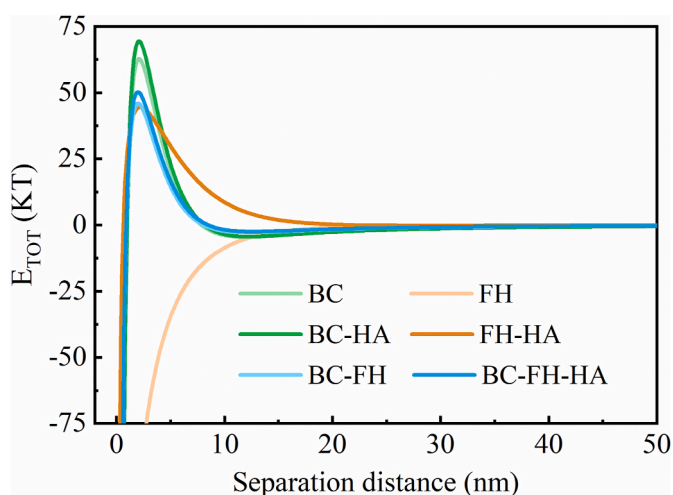


Fig. 5. The Derjaguin–Landau–Verwey–Overbeek interaction energies of the nanoparticles versus soil.

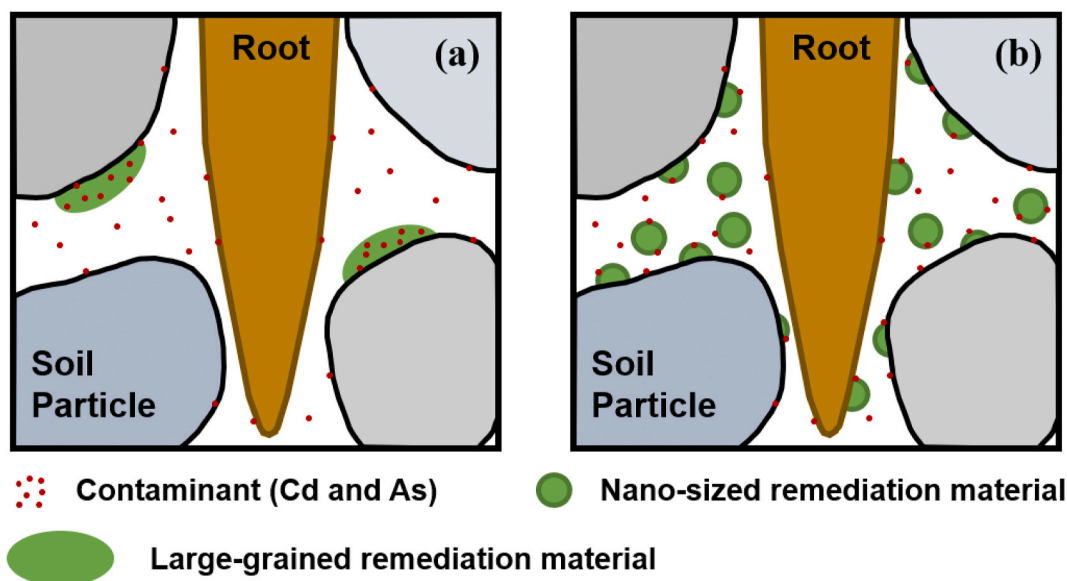


Fig. 6. Mechanism of nano-sized (a) and large-grained (b) remediation materials causing decreases in Cd and As uptake by the plant.

remediation materials lowered the Cd and As concentrations in the two types of monitored water more effectively than the nanomaterials. Despite this, the nano-sized treatments more effectively decreased the average Cd and As concentrations in the pak choi by more than 10% relative to the concentrations achieved by the large-grained treatments. The presence of HA had no obvious influence on the nano-sized material remediation process or on the uptake of Cd and As by the plants. The DLVO results also suggested that some of the BC_N , FH_N , and $BC-FH_N$ were immobilized in the topsoil, where they effectively adsorbed heavy metal(loid)s from the rhizosphere soil. The rest of the nano-sized materials were dispersed throughout the rhizosphere soil pores, blocking the uptake of Cd and As by the pak choi roots. Although the nano-material doses were less than one-fortieth of those of the large-grained materials in this study, the changes in the plant rhizosphere microenvironments caused by the nanomaterials decreased the toxicity transfer risk from the soil to the plant. In this study, nanoscale materials have been shown to be potential solutions for remediating agricultural soil co-contaminated by Cd and As and for alleviating the hazards these toxic species pose to plants and humans.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.chemosphere.2021.130731>.

CRedit author statement

Xiaoxue Ouyang: Writing – original draft, Methodology, Formal analysis, Visualization. Jie Ma: Conceptualization, Writing – review & editing, Funding acquisition. Pan Li: Writing – review & editing. Yali

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